

# INVERSES

**Definition:** The inverse of an  $n \times n$  matrix "A" is another  $n \times n$  matrix "C" such that:

$$A \cdot C = I_n \quad \text{AND} \quad C \cdot A = I_n$$

(often, you'll see "C" written as " $A^{-1}$ ")

On the homework assignment due today, you saw:

**Theorem:** Let  $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ . If  $ad - bc \neq 0$ , then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

If  $ad - bc = 0$ , then A is not invertible.

The quantity " $ad - bc$ " is the determinant of A

$$\det A = ad - bc.$$

**Ex** Find the inverses of the matrices

Ⓐ  $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \longrightarrow$  use above theorem!  
 $\det B = 4 - 6 = -2.$

so,  $B^{-1} = \frac{-1}{2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}.$

And we can check that our calculation is correct:

$$B \cdot B^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} -2+3 & 0 \\ 0 & 3-2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = B^{-1} \cdot B.$$

Ⓑ  $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ .  $\det I_2 = 1$ , so  $I_2^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ . ( $I_2$  is its own inverse!)

**Ex** Use inverses to solve the system

$$-x + 2y = 4$$

$$2x - 3y = 1$$

This is the same as writing:

$$\begin{bmatrix} -1 & 2 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

to solve for  $\begin{bmatrix} x \\ y \end{bmatrix}$ , instead of row reducing we can multiply each side by the inverse of  $\begin{bmatrix} -1 & 2 \\ 2 & -3 \end{bmatrix}$ !

$$\begin{bmatrix} -1 & 2 \\ 2 & -3 \end{bmatrix}^{-1} = \frac{1}{3-4} \begin{bmatrix} -3 & -2 \\ -2 & -1 \end{bmatrix} = -1 \begin{bmatrix} -3 & -2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$$

given..

March 12 2026

$$\begin{bmatrix} -1 & 2 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

we can multiply both sides of the equation on the left by the inverse matrix.

$$\underbrace{\begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}}_{I_2!} \begin{bmatrix} -1 & 2 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} \quad \text{then} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$\text{so } \begin{bmatrix} x \\ y \end{bmatrix} = 4 \begin{bmatrix} 3 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 14 \\ 9 \end{bmatrix}.$$

which gives our final solution:

$$x = 14 \\ y = 9.$$

## KERNELS

We've had some confusion computing kernels in the past. Before our exam, I want to make sure we can do it!

Recall:

**Definition** The kernel or null space of a linear transformation  $T(x) = A \cdot x$  is the set of  $x$  (in the domain) such that  $Ax = 0$ .

**Ex** Find the kernel of the linear transformation  $T(x) = Ax$ , where

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

First: set up augmented matrix, then row reduce.

$$\begin{array}{l} R1 \\ R2 \\ R3 \end{array} \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{array} \right] \xrightarrow{R2 \rightarrow R2 - 4R1} \begin{array}{l} R1 \\ R2 \\ R3 \end{array} \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & -3 & -6 & 0 \\ 7 & 8 & 9 & 0 \end{array} \right] \xrightarrow{R2 \rightarrow -\frac{1}{3}R2} \begin{array}{l} R1 \\ R2 \\ R3 \end{array} \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 7 & 8 & 9 & 0 \end{array} \right] \xrightarrow{R3 \rightarrow R3 - 7R1} \begin{array}{l} R1 \\ R2 \\ R3 \end{array} \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & -6 & -12 & 0 \end{array} \right] \xrightarrow{R3 \rightarrow 6R2 + R3} \begin{array}{l} R1 \\ R2 \\ R3 \end{array} \left[ \begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R1 \rightarrow -2R2 + R1} \begin{array}{l} R1 \\ R2 \\ R3 \end{array} \left[ \begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

to actually write down the kernel:

$$\begin{array}{rcl} x + 0y - z & = & 0 \\ y + 2z & = & 0 \\ 0 & = & 0 \end{array}$$

$$\begin{array}{l} x = z \\ y = -2z \\ z = z \end{array}$$

$$\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

This is as far as we can reduce!

$$\text{so } \text{kernel}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \right\}$$